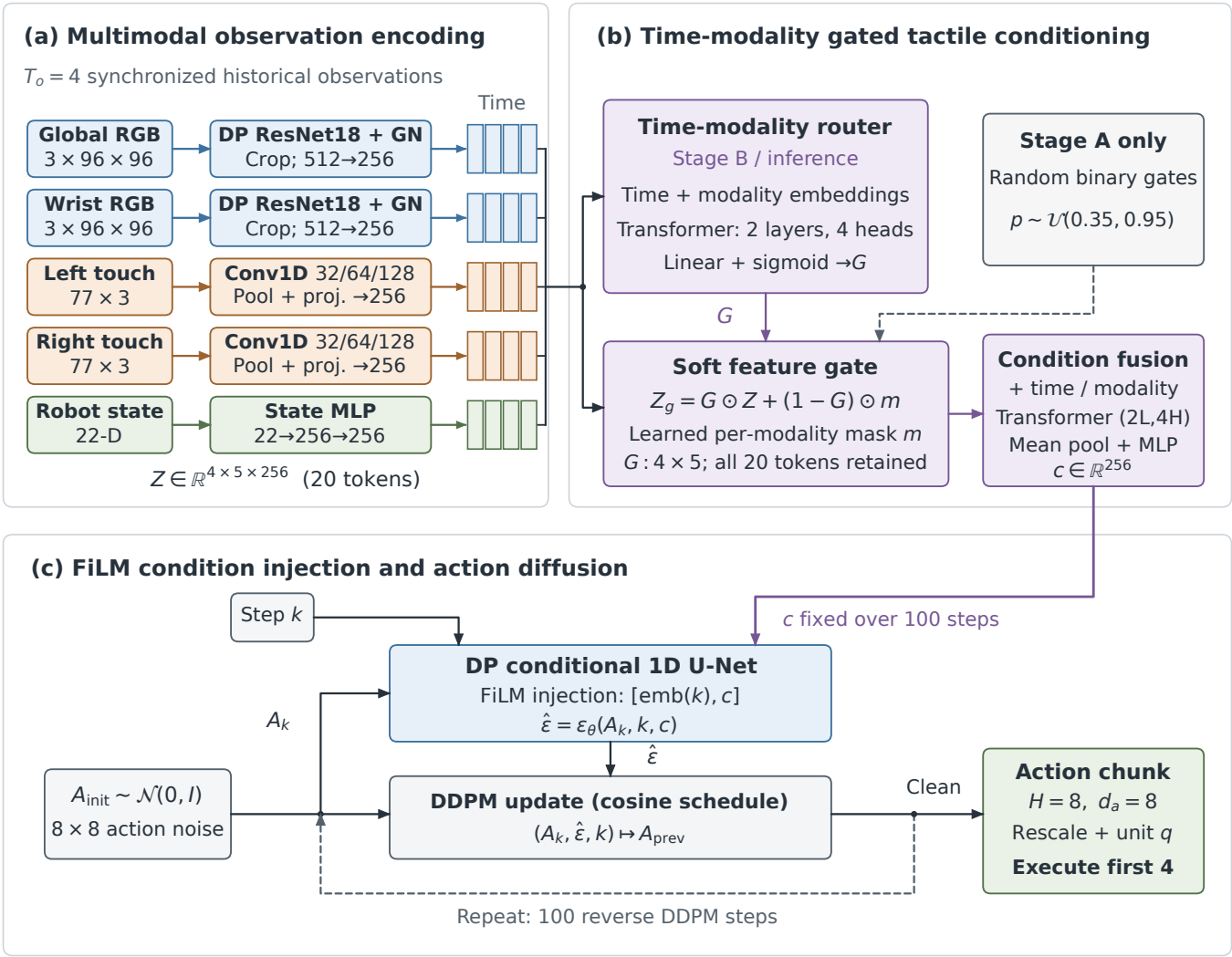
(a) Multimodal observation encoding
$T_o = 4$ synchronized historical observations
Time
Global RGB
3 × 96 × 96
DP ResNet18 + GN
Crop; 512→256
Wrist RGB
3 × 96 × 96
DP ResNet18 + GN
Crop; 512→256
Left touch
77 × 3
Conv1D 32/64/128
Pool + proj. →256
Right touch
77 × 3
Conv1D 32/64/128
Pool + proj. →256
Robot state
22-D
State MLP
22→256→256
$Z \in \mathbb{R}^{4 \times 5 \times 256}$ (20 tokens)
(b) Time-modality gated tactile conditioning
Time-modality router
Stage B / inference
Time + modality embeddings
Transformer: 2 layers, 4 heads
Linear + sigmoid →G
Stage A only
Random binary gates
$p \sim \mathcal{U}(0.35, 0.95)$
G
Soft feature gate
$Z_g = G \odot Z + (1 - G) \odot m$
Learned per-modality mask m
$G : 4 \times 5$; all 20 tokens retained
Condition fusion
+ time / modality
Transformer (2L,4H)
Mean pool + MLP
$c \in \mathbb{R}^{256}$
(c) FiLM condition injection and action diffusion
Step k
c fixed over 100 steps
DP conditional 1D U-Net
FiLM injection: [emb(k), c]
$\hat{\varepsilon} = \varepsilon_\theta(A_k, k, c)$
A_k
$\hat{\varepsilon}$
$A_{\text{init}} \sim \mathcal{N}(0, I)$
8 × 8 action noise
DDPM update (cosine schedule)
$(A_k, \hat{\varepsilon}, k) \mapsto A_{\text{prev}}$
Clean
Action chunk
$H = 8,\ d_a = 8$
Rescale + unit q
Execute first 4
Repeat: 100 reverse DDPM steps